2	**(a)**	<u>constant</u> rate of increase in velocity/acceleration from $t = 0$ to $t = 8\,\text{s}$	B1	
		<u>constant</u> deceleration from $t = 8\,\text{s}$ to $t = 16\,\text{s}$ or constant rate of increase in velocity in the opposite direction from $t = 10\,\text{s}$ to $t = 16\,\text{s}$	B1	[2]
	(b)	**(i)** area under lines to 10 s	C1	
		(displacement =) $(5.0 \times 8.0) / 2 + (5.0 \times 2.0) / 2 = 25\,\text{m}$ or $\frac{1}{2}(10.0 \times 5.0) = 25\,\text{m}$	A1	[2]
		(ii) $a = (v - u) / t$ or gradient of line	C1	
		$= (-15.0 - 5.0) / 8.0$		
		$= (-)\ 2.5\,\text{m}\,\text{s}^{-2}$	A1	[2]
		(iii) $\text{KE} = \frac{1}{2} m v^2$	C1	
		$= 0.5 \times 0.4 \times (15.0)^2 = 45\,\text{J}$	A1	[2]
	(c)	(distance =) $25\,(\text{m})\ (= ut + \frac{1}{2} a t^2) = 0 + \frac{1}{2} \times 2.5 \times t^2$	C1	
		($t = 4.5\ (4.47)\,\text{s}$ therefore) time to return $= 14.5\,\text{s}$	A1	[2]